1. a) $a^2 - b^2 > 0$

$a = -2 \quad b = 1$

$4 - 1 > 0$

$-2 - 1 = -3$

-3 is not greater than zero

b) $\frac{7}{7} = 1$

c) $\sqrt{3} \times 3\sqrt{3} = 9$

d) $\cos 90 - |x| = \sin x$

$x = -90$

$|x| = 90$

$\cos(90 - 90) = 1$

$\sin(-90) = -1$

2. a) True $6 = 2(3)$

So divisible by 2

b) $(3^{\wedge} + 2)$ False

$3^5 + 2 = 245$

$245 \div 5 = 49$

c) $\sqrt{n}$ False

$\sqrt{4} = 2$

d) $\frac{a}{b} \quad \frac{b}{c} \quad \frac{a}{c}$

$\frac{a}{c} = \frac{a}{b} \times \frac{b}{c}$

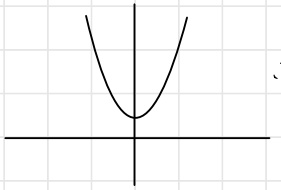
$x^2 - 8x + 1770 \quad x \in \mathbb{R}$

$a = 1$

$b = -8$

$c = 17$

$(-8)^2 - 4 \times 1 \times 17 = -4 \therefore$ no roots



$\therefore x^2 - 8x + 1770$

$(n+3)^2 > n^2$

$n = 4 \quad (4+3)^2 > 4^2$

$n = -2$

$(-2+3)^2 < (-3)^2$

$1 < 2$

Sometimes true

2. $\sqrt{xy} \leq \frac{x+y}{2}$

$(x-y)^2 \geq 0$

$x^2 - 2xy + y^2 \geq 0$

$(2\sqrt{xy})^2 \leq (x+y)^2$

$4xy \leq (x+y)^2$

$\therefore 2\sqrt{xy} \leq x+y$

$\sqrt{xy} \leq \frac{x+y}{2}$

b) $x=-3 \quad y=-4$

$\sqrt{12} \leq \frac{-7}{2}$

$3.46 \leq -3.5$

not true

3. $n \in N$

n^3+2 not divisible by 8

If n = odd $\therefore n^3$ = odd

n^3+2 not divisible by 8

$n = 2m$

$(2m)^3+2 = 8m^3+2$

$8m^3+2$ not divisible by 8

4. a) $\frac{4a}{b} + \frac{b}{a} \geq 4$

$\frac{4a^2}{ab} + \frac{b^2}{ab} = \frac{4a^2+b^2}{ab} \geq 4$

$4a^2+b^2 \geq 4ab$

$4(a^2+b^2) \geq 4(ab)$

$a^2+b^2 \geq ab$

$a=-7 \quad b=4$

$\frac{4(-7)}{4} + \frac{4}{-7} = \frac{-53}{7}$

13.05.23

1. $\int_a^{2a}\left(\frac{t+1}{t}\right)dt = \ln 7$

$t\sqrt{t+1}^{\wedge}\frac{1}{t}$

$\int_a^{2a}\left(1+\frac{1}{t}\right)dt = \ln 7$

$\left[t+\ln|t|\right]_a^{2a} = \ln 7$

$(2a+\ln 2a)-(a+\ln a)$

$2a-a=a$

$\ln 2a - \ln a = \ln\left(\frac{2a}{a}\right) = \ln 2$

$a+\ln 2 = \ln 7$

$a = \frac{\ln 7}{\ln 2} = \ln 7_2 \quad k = 7_2$

b) $\int_k^{2k}\frac{2}{(2x-k)^2}dx$

$2(2x-k)^{-2}$

$2\div -1\div 2$

$-1(2x-k)^{-1}$

$\left[\frac{-1}{(2x-k)}\right]_k^{2k}$

$\left(\frac{-1}{(4k-k)}\right)-\left(\frac{-1}{2k-k}\right) = \frac{-1}{3k}+\frac{1}{k}$

$\frac{-1}{3k}+\frac{3}{3k} = \frac{2}{3k}$

2. a) $\int_k^{3k}\frac{2}{(3x-k)}dx$

$\left[\frac{2}{3}\ln|3x-k|\right]_k^{3k}$

$\frac{2}{3}\ln(9k-k) - \frac{2}{3}\ln(3k-k)$

$\frac{2}{3}\ln\left(\frac{8k}{2k}\right) = \frac{2}{3}\ln\frac{8}{2}$

$= 0.9242$

3. a) $y = \frac{x^2\ln x}{3} - 2x + 5 \quad x>0$

x	1	1.5	2	2.5	3
y	3	2.3041	1.9242	1.9089	2.2958

(x,y) — abscissa, ordinate

$h = \frac{b-a}{n} = \frac{3-1}{4} = 0.5$

$\frac{1}{2}h\left(y_1+y_9+2(y_2+y_3+y_4+y_5+y_6\ldots)\right.$

$\frac{1}{2}\left(3+2.2958+2(2.3\ldots+1.9\ldots+1.9\ldots)\right)$

$= 8.7851 \times 1_2 = 4.39255$

b) Increase number of strips

c) $\int_1^3 \frac{x^2\ln x}{3} - 2x + 5\,dx$

c) $\int_1^3 \frac{x^2 \ln x}{3} - 2x + 5 \, dx$

$\frac{1}{3}\int_1^3 x^2 \ln x \, dx + \int_1^3 -2x + 5 \, dx$

$v = \ln x \qquad v = \frac{x^3}{3}$

$\frac{dv}{dx} = \frac{1}{x} \qquad \frac{dv}{dx} = x^2$

$I = uv - \int v \frac{dv}{dx} \, dx$

$\ln x \frac{x^3}{3} - \int \frac{x^3}{3} \frac{1}{x} \, dx$

$\frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$

$\frac{x^3}{3} \ln x - \frac{1}{9} x^3$

$\left[\frac{x^3}{9} \ln x - \frac{x^3}{27} - x^2 + 5x \right]_1^3$

$\left[\frac{x^3}{9} \ln x - \frac{x^3}{27} - x^2 + 5x \right]_1^3$

$\frac{3^3}{9} \ln 3 - \frac{3^3}{27} - 3^2 + 5(3) =$

$\left(3 \ln 3 - 1 - 9 + 15\right) - \left(\frac{1}{9} \ln 1 - \frac{1}{27} - 1 + 5\right)$

$(3 \ln 3 + 5) - \left(\frac{107}{27}\right) = 3 \ln 3 + \frac{28}{27} = \ln 3^3 + \frac{28}{27}$

$\ln 27 + \frac{28}{27}$

4. $\int_0^2 2x\sqrt{x+2}\,dx = \frac{32}{15}(2+\sqrt{2})$

$\frac{64}{15} - \left(\frac{-32}{15}\sqrt{2}\right) =$

$\frac{64}{15} + \frac{32}{15}\sqrt{2} = \frac{32}{15}(2+\sqrt{2})$

$v = x+2$ $\quad v = 2+2 = 4$

$x = v-2$ $\quad v = 0+2 = 2$

$\frac{dx}{dv} = 1$

$dx = dv$

$\int_2^4 (2v-4) \times v^{1/2}\,dv$

$\int_2^4 2v^{3/2} - 4v^{1/2}\,dv$

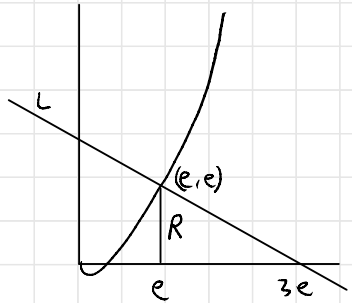
$\left[\frac{4}{5}v^{5/2} - \frac{8}{3}v^{3/2}\right]_2^4$

$\left(\frac{4}{5}4^{5/2} - \frac{8}{3}4^{3/2}\right) - \left(\frac{4}{5}2^{5/2} - \frac{8}{3}2^{3/2}\right)$

$\left(\frac{64}{15}\right) - \left(4\sqrt{2} \times \frac{4}{5} - \frac{8}{3} \times 2\sqrt{2}\right)$

$\left(\frac{64}{15}\right) - \left(\frac{16}{5}\sqrt{2} - \frac{16}{3}\sqrt{2}\right) = \left(\frac{64}{15}\right) - \left(\frac{48\sqrt{2}}{15} - \frac{80}{15}\sqrt{2}\right)$

5.



$y = x \ln x$

L = normal to C at point e, e

$\frac{dy}{dx} = UV' + VU'$

$= x \frac{1}{x} + \ln x (1)$

$\frac{x}{x} + \ln x = 1 + \ln x$

$\frac{dy}{dx}$ when $x = e$

$1 + \ln e = 2$

$y = 2x + c$

$y = -\frac{1}{2}x + c$

$y - y_1 = m(x - x_1)$

$y - e = -\frac{1}{2}(x - e)$

$y = -\frac{1}{2}x + \frac{1}{2}e + e$

$-\frac{1}{2}x + \frac{1}{2}e + e = 0$

$\frac{3}{2}e = \frac{1}{2}x$

$\frac{6}{2}e = x$

$3e = x$

$y = x \ln x = 0$

$x = 0$

$\ln x = 0 \quad x = 1$

$\int_1^e x \ln x \, dx$

$U = \ln x \qquad V = \frac{x^2}{2}$

$\frac{du}{dx} = \frac{1}{x} \qquad \frac{dv}{dx} = x$

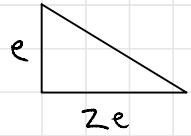
$\frac{x^2}{2}\ln x - \int \frac{x^2}{2}\frac{1}{x}\,dx$

$\frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx$

$\left[\frac{x^2}{2}\ln x - \frac{1}{4}x^2\right]_1^e$

$\left(\frac{e^2}{2}\ln e - \frac{1}{4}e^2\right) - \left(-\frac{1}{4}\right)$

$\frac{1}{2}e^2 - \frac{1}{4}e^2 + \frac{1}{4}$



$\frac{2e^2}{2} = e^2$

$\frac{5}{4}e^2 + \frac{1}{4}$

100–200	100
300–400	250
500–700	600
800–1100	50

Freq dens = $\frac{\text{Freq}}{\text{Class width}}$

$\frac{250}{100} = \frac{25}{10} = 2.5$



$\bar{x} = 85.5$

$n = 7$

$\frac{7(85.5) + x}{8} = 84.9$

$7(85) + x = 679.2$

$598.5 + x = 679.2$

$x = 80.7$

$n = 100$

$\Sigma(x-20) = 125$

$\Sigma(x-20)^2 = 221$

$\frac{100(\bar{x}-20)}{100} = 125$

$\bar{x} - 20 = 125$

$\bar{x} = 105$

$1 + \cos 2x = 1.4$

$x = 33.2$

$\cos 2x = 0.4$

$\cos\theta = 0.4$

$\cos^{-1} 0.4 = \theta$

$\theta = 66.422$

$y = ax^2 + 6x + c$

$x = 0 \quad y = 10 \quad 10 = c$

minimum point $-1, 7$

$y = ax^2 + 6x + 10$

$\frac{dy}{dx} = 2ax + 6 = 0$

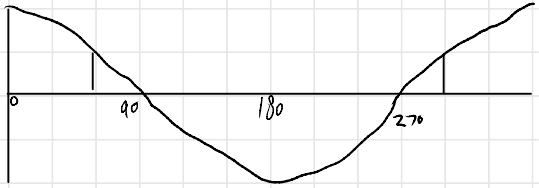
$2a = -6$

$a = -3$

$x = -1$

$-a = -3$

$a = 3$



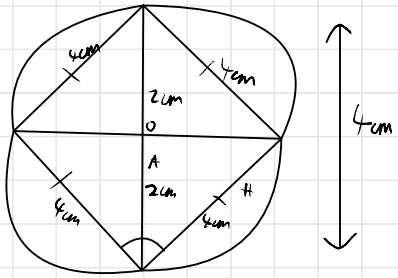
Freq dense = $\frac{\text{Freq}}{\text{class width}}$

$\frac{84}{1} = 84 \div x = 1$

12

$\frac{15}{4} \div 12 = 0.3125$

1.



$\theta = 2\left(\cos^{-1}\frac{2}{4}\right) = \frac{2\pi}{3}$

$A = \frac{1}{2} r^2 \theta$

$\frac{1}{2} \times 4^2 \times \frac{2\pi}{3} = \frac{16}{3}\pi$

$A = 4 \times \frac{1}{2} \times 2 \times \sqrt{4^2 - 2^2}$

$= 8\sqrt{3}$

$\frac{16}{3}\pi - 8\sqrt{3}$

$\frac{32\pi}{3} - 16\sqrt{3} = \frac{2}{3}\left(\frac{16}{3}\pi - 24\sqrt{3}\right)$

2. $\sin\theta = \theta \quad \cos\theta = 1 - \frac{1}{2}\theta^2 \quad \tan\theta = \theta$

$\frac{1 + \sin\theta + \tan 2\theta}{2\cos 3\theta - 1} = \frac{1}{1 - 3\theta}$

$\frac{1 + \theta + 2\theta}{2\left(1 - \frac{(3\theta)^2}{2}\right) - 1} = \frac{1 + 3\theta}{2 - 9\theta^2 - 1} = \frac{1 + 3\theta}{1 - 9\theta^2}$

$\frac{1 + 3\theta}{(1 + 3\theta)(1 - 3\theta)} = \frac{1}{1 - 3\theta}$

$\theta = \text{small} \quad \frac{1}{1} = 1$

3. $\frac{\tan x - \sec x}{1 - \sin x} = \frac{\frac{\sin x}{\cos x} - \frac{1}{\cos x}}{1 - \sin x} = \frac{\sin x - 1}{\cos x} \times \frac{1}{1 - \sin x}$

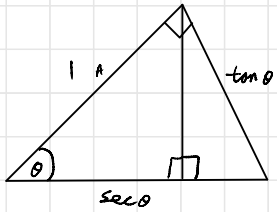
$\frac{\sin x - 1}{\cos x(1 - \sin x)} = \frac{-1(1 - \sin x)}{\cos x(1 - \sin x)} = \frac{-1}{\cos x} = -\sec x$

$-\sec x = \sqrt{2}$

$\frac{1}{\cos x} = -\sqrt{2}$

$\cos x = \frac{-1}{\sqrt{2}} \qquad x = \frac{3}{4}\pi, \frac{5}{4}\pi$

4.



$\tan\theta = \frac{BC}{1}$ $\cos\theta = \frac{1}{AC}$

$AC = \sec\theta$

$AB^2 + BC^2 = AC^2$

$1 + \tan^2\theta = \sec^2\theta$

5. Solve $6\sin(\theta+60) = 8\sqrt{3}\cos\theta$

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$6(\sin\theta\cos 60 + \cos\theta\sin 60) = 8\sqrt{3}\cos\theta$

$3\sin\theta + 3\sqrt{3}\cos\theta = 8\sqrt{3}\cos\theta$

$3\sin\theta = 5\sqrt{3}\cos\theta$

$3\tan\theta = 5\sqrt{3}$

$\tan\theta = \frac{5\sqrt{3}}{3}$ $\theta = 70.9, 250.9$

6. $\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= 2\cos^2\theta - 1$

$= 1 - 2\sin^2\theta$

$(\sin 3\theta + \cos 3\theta)^2 = 1 + \sin 6\theta$

$\sin^2 3\theta + 2\sin 3\theta\cos 3\theta + \cos^2 3\theta$

$1 + 2\sin 3\theta\cos 3\theta$

$1 + \sin 6\theta$

$(\sin 3\theta + \cos 3\theta)^2 = \frac{2+\sqrt{2}}{2}$

$\sin 6\theta = \frac{2}{2} + \frac{\sqrt{2}}{2} - 1 = \frac{\sqrt{2}}{2}$

$6\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5}{4}\pi, \frac{9}{4}\pi, \frac{11}{4}\pi$

7. $5\cos\theta - 8\sin\theta = R\cos(\theta+\alpha)$

$R\cos\theta\cos\alpha = 5\cos\theta$

$+ R\sin\theta\sin\alpha = +8\sin\theta$

$\tan^{-1}\frac{8}{5} = 58^\circ$ or 1.0122^c

$\sqrt{5^2+8^2} = \sqrt{89}$

$T = 1100 + \sqrt{89}\cos\left(\frac{x}{3} + 1.0122\right)$

$T = 1100 + \sqrt{89} = 1109.43$

$\frac{x}{3} + 1.0122 = 2\pi$

$x = 3(2\pi - 1.0122)$

$x = 15.81$

Statistics

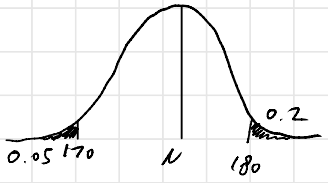
1. $z = 1\frac{1}{3}$
$z = 1.2$

2. $m \sim N(900, 100^2)$
$P(m < 850) = 0.3085$

$771 < x < 1028$.

3. $X \sim B(200, 0.54)$ $P(X > 100)$
$Y \sim N(108, 49.68)$ $P(Y > 100.5) = 0.8564$

4.



$(P(X > 175))^3 = 0.2806$

$A = 0.05$ $Z = -1.6449$

$-1.6449 = \frac{170 - N}{\sigma}$ $= N - 1.6449\sigma = 170$

$A = 0.8$ $Z = 0.8416$ $N = 178.6$

$N + 0.8416\sigma = 180$ $\sigma = 4.02$

5. Large N
P = roughly 0.5
Have no batteries to sell
$X \sim B(300, 0.55)$

$H_0 : p = 0.55$ 5%.
$H_1 : p > 0.55$ onetailed
$P(X \geq 184)$

$Y \sim N(165, 74.25)$
$P(Y > 183.5) = 0.0159$
$0.0159 < 0.05$
reject H_0, Sufficient evidence to backup manufacturers claim at 5% level of significance.